IME: Working Principle

A Doppler spectral estimation method designed to identify reliable atmospheric peaks under noisy and low-SNR conditions.

Main Processing Stages

- Spectral Preprocessing
  - Remove DC components and ground clutter.
  - Apply 5-point smoothing and estimate the noise level.
- Adaptive Doppler Window
  - Restrict peak searching to a selected Doppler range.
  - Identify prospective peaks using their SNR.
- Profile Tracing
  - Connect candidate peaks across adjacent range bins.
  - Select the profile using velocity continuity and relative SNR.